MJOS

14 Din ka Offline Database Mission

PostgreSQL + Django ORM | MongoDB CRUD

14 din baad tumhara result kya hona chahiye?

- [ ] MJOS ke db_lab home page par PostgreSQL ka data ek saaf table mein nazar aaye.
- [ ] Table se Add, Read, Edit aur Delete buttons kaam karein.
- [ ] Naam se search ho aur ghalat/khali input par samajh aane wala paigham aaye.
- [ ] Tum khud bata sako ke Django form se data database mein kaise save aur wapas display hota hai.
- [ ] MongoDB ka Create, Read, Update, Delete (CRUD) ek alag practice database mein chal jaye.

Focus ka seedha rule

POSTGRESQL	MONGODB
Main practice. db_lab ke GUI aur Django ORM mein use karo.	Sirf alag practice. MJOS ke real features mein abhi connect nahi karna.

Daily rule

- Minimum 30 minutes; aam target 60-90 minutes. Har din sab kuch karne ki zaroorat nahi.
- Har din ek chhota result note karo: kya chala, kya error aya, kal kya karna hai.
- Din miss ho jaye to plan cancel nahi hota. Agley din wahi task se wapas shuru karo.
- Movies/series rest ke liye hain. Pehle daily minimum task; phir entertainment.

Important: Yeh learning roadmap hai, production release checklist nahi. Pehle working aur samajh; baad mein extra features.

Aakhir mein GUI kuch aisa hoga

ID	Name	Price	Quantity	Actions
1	Keyboard	3500	20	Edit | Delete
2	Mouse	1200	8	Edit | Delete

Table ke upar: [Search by name] [Add Product]

Sample layout hai. Asal rows PostgreSQL se aayengi; values hard-code nahi karni.

Internet band hone se pehle: setup check

Yeh list internet band hone se pehle complete karo, kyun ke baad mein package download ya server install karna mushkil ho sakta hai.

- [ ] MJOS ka backup/Git status check karo. Apni important unsaved files ka copy bhi rakho.
- [ ] Virtual environment activate karke python manage.py check chalao; errors hon to abhi solve karo.
- [ ] PostgreSQL service aur MJOS ka db_lab page local machine par khul raha ho.
- [ ] db_lab mein jo Product record pehle se hai, usay browser/shell mein dekh kar confirm karo.
- [ ] Venv mein psycopg2/psycopg2-binary aur pymongo packages available hain ya nahi check karo.
- [ ] MongoDB SERVER local PC par installed aur chalne ke qabil ho. Sirf pymongo package kaafi nahi.
- [ ] Agar MongoDB server installed nahi, to internet band hone se pehle setup verify karo; warna Mongo task ko force mat karna.
- [ ] Django settings mein MongoDB ko DATABASES mein add mat karo. PostgreSQL config ko stable rehne do.
- [ ] db_layer.py ka pasted connection code adhoora hai. Filhal isay views se connect mat karo aur is par depend mat karo.
- [ ] Product ke price field ka type model aur migrations mein check karo. Decimal value nazar aayi thi, is liye type andazay se change mat karna.

Offline ready? [ ] PostgreSQL works [ ] db_lab opens [ ] MongoDB server verified [ ] Backup ready

Din-ba-din roadmap

DIN 1 | Current app ko samjho

Maqsad: App ki files aur data ka safar samajhna.

- [ ] db_lab ka page kholo aur Product record dekho.
- [ ] Flow paper par likho: urls.py -> views.py -> models.py/database -> template.
- [ ] python manage.py check aur local server dobara verify karo.

Din ka saboot: Tum app ka data-flow apne alfaaz mein explain kar sako.

DIN 2 | Database se data parhna

Maqsad: ORM result aur asal record ka farq samajhna.

- [ ] Product.objects.all(), values(), get() aur filter() ko shell mein test karo.
- [ ] Samjho ke values() dictionaries deta hai; Manager par .keys() nahi chalta.
- [ ] Keyboard record ka id, name, price aur quantity verify karo.

Din ka saboot: Har query ka expected result pehle predict kar ke phir check karo.

DIN 3 | Search queries ka bunyadi logic

Maqsad: Zaroorat ka record dhoondhna aur result arrange karna.

- [ ] filter(), exclude(), order_by(), count() aur exists() ke examples chalao.
- [ ] SQL ke SELECT, WHERE aur ORDER BY ka matlab samjho.
- [ ] Alag test values use karo; original record ko be-wajah delete mat karo.

Din ka saboot: Kam az kam 5 queries ka result samjha sako.

DIN 4 | Shell mein CRUD practice

Maqsad: Code se record banana aur badalna samajhna.

- [ ] Ek naya test Product banao; usay read karo, update karo aur phir sirf us test record ko delete karo.
- [ ] Har operation ke baad database se result verify karo.
- [ ] Create, Read, Update, Delete ke apne lafzon mein 1-1 jumla likho.

Din ka saboot: Test record ka poora CRUD cycle complete ho.

Part A - db_lab ko GUI banana

Is hissa ka final result: browser mein table ho aur buttons asal PostgreSQL data ko badlein; sirf screen par temporary change na ho.

DIN 5 | Database ka data table par dikhao

Maqsad: Read operation ko GUI mein lana.

- [ ] Existing home template use karo; zaroorat par usay db_lab.html naam do, lekin duplicate template mat banao.
- [ ] View mein Django ORM se Product records lao aur template ko bhejo.
- [ ] Table columns rakho: ID, Name, Price, Quantity, Actions.
- [ ] Agar record na ho to “Abhi koi product nahi” jaisa message dikhao.

Din ka saboot: Refresh ke baad bhi table mein database ke records nazar aayen.

DIN 6 | Add Product form

Maqsad: Browser se naya record save karna.

- [ ] Add Product button aur chhota form banao: Name, Price, Quantity.
- [ ] Django form/ModelForm aur server-side validation use karo.
- [ ] POST request aur CSRF protection use karo; successful save ke baad user ko message do.
- [ ] Khali ya invalid input par error usi form ke paas nazar aaye.

Din ka saboot: Form se banaya product refresh ke baad bhi table mein rahe.

DIN 7 | Edit button

Maqsad: Maujooda record ko badalna.

- [ ] Har table row mein Edit button ho.
- [ ] Edit kholne par usi product ki existing values form mein pehle se bhari hon.
- [ ] Save karne par sirf selected product update ho; naya duplicate record na bane.
- [ ] Update ke baad success message aur updated table dikhni chahiye.

Din ka saboot: Ek product ka price/quantity change ho aur database mein bhi wahi change ho.

DIN 8 | Delete button

Maqsad: Record ko soch samajh kar delete karna.

- [ ] Har row mein Delete button ho aur confirmation poochi jaye.
- [ ] Delete ke liye POST request use karo; sirf link/GET kholne se record delete na ho.
- [ ] Delete ke baad message dikhao aur table refresh/update karo.
- [ ] Sirf apna test record delete karke database mein verify karo.

Din ka saboot: Selected record table aur database dono se remove ho.

GUI ka matlab: buttons aur forms kaam karein. Sirf button bana dena kaafi nahi; har button ke baad database result verify karna hai.

Part B - Queries, migrations aur relations

DIN 9 | Search aur table ko behtar banao

Maqsad: Record jaldi milna aur table readable hona.

- [ ] Name search box add karo; search ko GET query se chala sakte ho.
- [ ] Search ke sath no-results message aur clear-search option rakho.
- [ ] Price ya name ke hisaab se sorting test karo.
- [ ] Har Add/Edit/Delete action ko dobara test karo; search se CRUD na tootay.

Din ka saboot: Search aur tamam buttons working hon. Pagination optional hai, lazmi nahi.

DIN 10 | Migrations samjho

Maqsad: Model change aur database table update ka farq samajhna.

- [ ] db_lab/migrations folder mein existing migration files dekho.
- [ ] makemigrations ka matlab: model change ka migration record banana.
- [ ] migrate ka matlab: pending migration database par apply karna.
- [ ] sqlmigrate se dekho Django SQL mein kya tabdeeli karega.
- [ ] Pehle backup aur current price field ka type verify kiye baghair field type change ya database reset mat karo.

Din ka saboot: Makemigrations, migrate aur sqlmigrate ka farq explain kar sako.

DIN 11 | Category aur Product ka relation

Maqsad: Do related tables ka basic idea samajhna.

- [ ] Pehle paper par samjho: ek Category ke andar kai Products ho sakte hain.
- [ ] Sirf tab Category model + Product ForeignKey ka experiment karo jab GUI CRUD stable ho.
- [ ] Migration banao/apply karo aur shell mein category se related products dekho.
- [ ] Agar main CRUD abhi incomplete hai, is task ko skip karke CRUD bugs fix karo.

Din ka saboot: ForeignKey ka basic matlab samajh aaye; core CRUD ko risk mein na dalo.

Part C - MongoDB sirf alag practice

MongoDB ke liye MJOS ka main settings.py ya accounts/projects ko change nahi karna. Alag practice database aur apne test documents use karo.

DIN 12 | MongoDB: connection, Create aur Read

Maqsad: Document database ke sath pehli practical exercise.

- [ ] MongoDB server local par chal raha hai, yeh confirm karo. PyMongo package akela server nahi hota.
- [ ] Alag practice database aur collection rakho, misaal: mj_offline_lab / practice_products.
- [ ] Python/PyMongo se ek test document insert karo; phir find se records read karo.
- [ ] Document mein name, price aur quantity rakho. PostgreSQL se automatic copy/sync nahi karni.

Din ka saboot: MongoDB mein apna test document insert aur read ho.

DIN 13 | MongoDB: Update aur Delete

Maqsad: MongoDB ka poora CRUD cycle complete karna.

- [ ] Apne test document ko filter ke zariye dhoondo.
- [ ] Update operation se sirf ek field badlo aur result dobara read karo.
- [ ] Apne test document ko delete karo; phir verify karo ke woh nahi raha.
- [ ] Find/filter, sort aur count ka basic practice karo. Sirf practice collection mein changes karo.

Din ka saboot: MongoDB ka Create, Read, Update, Delete chaaron successful hon.

DIN 14 | Final test - bina AI ke

Maqsad: Apni learning ko khud verify karna.

- [ ] Browser se naya Product add karo, page refresh karo, phir usay search karo.
- [ ] Usi product ko Edit karo, phir Delete karo. Har step database mein verify karo.
- [ ] MongoDB mein ek alag test document par CRUD dobara chalao.
- [ ] Notebook/OFFLINE_LOG.md mein likho: PostgreSQL kis kaam aa raha hai? MongoDB kis kaam aa raha hai? Dono ko abhi kyun integrate nahi kar rahe?
- [ ] Agar koi feature nahi chala, usay clearly note karo. Fake completion tick mat lagao.

Din ka saboot: PostgreSQL GUI CRUD aur separate MongoDB CRUD ka practical proof ho.

14 din baad final checklist

- [ ] db_lab home page par asal Product rows table mein dikh rahi hain.
- [ ] Add form record save karta hai; invalid input par error deta hai.
- [ ] Edit selected record ko update karta hai, duplicate create nahi karta.
- [ ] Delete confirmation ke baad selected record delete hota hai; GET se delete nahi hota.
- [ ] Search kaam karti hai; empty result par message aata hai.
- [ ] Django ORM/PostgreSQL aur MongoDB/PyMongo alag-alag practice hue hain.
- [ ] MongoDB ko MJOS ke real modules mein abhi integrate nahi kiya.
- [ ] Project ka backup aur daily learning log maujood hai.

AAKHRI RULE: PostgreSQL mein depth. MongoDB mein fundamentals + CRUD. Dono ko ek sath connect karna is 14 din ka goal nahi. Goal yeh hai ke tum apna database-backed GUI khud samajh kar bana sako.